Mohamed Bilel Khadhraoui Software Engineer

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EDUCATION

Master of Engineering in Computer Science 09/2020 – 09/2025 | Charguia, Tunisia
National Institute of Applied Sciences and Technologies
Control Theory ● Electronics ● Algebra ● Calculus ● Assembly ● UML ● OOP ● Algorithms ● Linux

WORK EXPERIENCE

Software Engineer - E-Mobility, *Watt&Well* 03/2025 – Present | Massy, France
- Architected and implemented a modular Go CANopen gateway for heterogeneous third-party power units, using interfaces to decouple the Gateway Manager from vendor-specific drivers and enable integration of new power-unit types without modifying core gateway logic.
- Designed a configuration-driven gateway architecture supporting flexible electrical configurations and grouping multiple external power units behind virtual power units; implemented concurrent measurement handling with mutex-protected shared state and isolated vendor communication through dedicated drivers.
- Implemented fault isolation and communication monitoring for external power units, propagating vendor-specific faults to virtual power units and exposing communication failures to the Charging Station Manager while allowing unaffected units to remain operational.
- Designed and implemented an independent Go metering process integrated with the Charging Station Manager through ZeroMQ, using interface-based meter abstractions to support multiple meter implementations while decoupling data collection from its consumers.
- Defined the inter-process communication specification and implemented typed JSON/OCMF decoding to transform meter data into the system's internal representation, including transaction-related data and measurements subsequently exposed through HTTP endpoints.
- Designed metering-process availability and communication-state handling, including CSM disconnected states and meter ping/timeout mechanisms for detecting unreachable meters.
- Defined technical specifications in collaboration with commercial engineers based on client requirements, coordinated interfaces with CSM engineers, and specified test-bench modules with technicians to ensure smooth system integration.
- Established automated GitLab CI pipelines covering static analysis/linting, unit tests, E2E/integration tests, compilation, documentation generation, and packaging; built E2E tests exercising CSM, the metering process, and mocked meters over virtual serial interfaces.

Embedded Linux Intern, *STMicroelectronics* 07/2024 – 09/2024 | Tunis, Tunisia
- Conducted an in-depth exploration of the OpenSTLinux distribution and boot chain of MP135 microprocessor; U-Boot, TF-A, OP-TEE, Kernel
- Crafted a high-performance kernel module for the Grove WiFi V1/V2 module using the TTY framework for serial communication (UART).
- Designed and implemented a Command Line Interface (CLI) in C, delivering efficient abstraction and control of the kernel module.
- Authored Yocto recipes for Device Tree configuration and cross-compilation, ensuring seamless deployment on the MP135 Microprocessor.

Embedded Software Engineer, *Shanon Technologies* 07/2023 – 01/2024 | Toulouse, France
- Benchmarked ShanonDSPWizard through complex block diagram implementation, performance evaluation, and live demonstration.
- Developed a cross-platform Software Abstraction Layer for STM32H723 FMAC Peripheral, enhancing portability and IIR/FIR filters.
- Designed a cross-platform middleware for CORDIC and CRC, providing APIs for configuration, manipulation with a technical documentation.
- Expanded the STM32H723 CORDIC's and FMAC's function range by implementing a recursion-based approach for mathematical decomposition to address the fixed-point system.

PROJECTS

Orca Project 11/2025 – 08/2026
- Reduced Linux kernel image size by 96% (from 450 MB to 15 MB) by stripping unnecessary modules and retaining project-specific drivers.
- Accelerated boot time by 2× through optimization of U-Boot env variables and implementation of a customized Fastboot command sequence.
- Automated reproducible builds by version-locking all required meta-layers in a KAS configuration, leveraging KAS-Container for builds.

Linux From Scratch 12/2025 – 01/2026
- Bootstrapped a complete cross-toolchain from source, starting with Binutils (linker and assembler) and a two-stage GCC compiler.
- Isolated the final filesystem build process inside a chrooted environment to ensure clean, reproducible, and build-independent packages.
- Built all essential utilities and packages (such as coreutils, diffutils, and Python 3) from source using a third-stage native toolchain.
- Compiled the GRUB bootloader and configured the Linux kernel to generate a UEFI-bootable image.

International Robotics Competition - Eurobot 11/2022 – 06/2023
- Realized autonomous mechanical systems for advanced automation; 3 DoF Robotic Arm, Rolling base.
- Optimized Dijkstra-based path planner using 2D geometric model of the robot's environment, achieving 4× faster computation-to-travel time.
- Integrated a state machine within the ROS environment using Python 3, enabling seamless mechanical synchronization.
- Formulated a control algorithm implemented on an STM32H745 MCU, employing odometry and PID control for enhanced maneuverability.
- Fine-tuned 12 parameters of the PID control algorithm to achieve an error margin of less than 1 mm from the desired coordinates.

SKILLS

Computational Thinking: Composition/Debugging/Abstraction | **Embedded Peripherals:** GPIO/TIMER/ADC/DAC/CORDIC/FMAC/CRC | **Microcontrollers:** STM32/ATmega/ESP32 | **Software:** CubeIDE/CubeProgrammer | **Programming:** Embedded C/Go/Bash/Python | **Test Equipment:** Logic Analyzer/Oscilloscope | **Microprocessors:** Raspberry Pi/STM32MP | **Version Control System:** Git/Github | **Communication Protocols:** UART/I2C/SPI/CAN/CANopen | **Embedded Linux:** Yocto/OpenSTLinux/Makefile/Cross Compilation/Kernel Dev

ORGANIZATIONS

TechMag, *Digital Creator* 01/2024 – 03/2025
- Produced high-quality digital content on advanced deeptech topics, engaging an audience of over 10,000 viewers on social platforms.
- Led the MagTech IT Segment live, delivered tech insights to over 5000 listeners and produced engaging content with over 9000 interactions.

IEEE RAS INSAT Student Branch, *Technical Trainer* 09/2022 – present
- Spearheaded more than 5 workshops; Embedded Linux, STM32 in collaboration with 3 different institutes, hosting more than 30 participants each.

LANGUAGES

• French | C1 • English | C1 • Arabic | Native Language • German | A1